

PRACTICE EXAM 5 OF 5 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 5

Twenty questions weighted towards monitoring, migration, hybrid and identity.

QUESTIONS 20 WEIGHTING 30/26/24/20 PASS MARK 14 of 20

A consumer application will have around five million registered users who sign in with email or a social provider. Users must also upload files directly from the browser to a private S3 bucket, each restricted to their own folder.

No file may pass through an application server.

Which combination meets these requirements? (Choose two.)

- ☐ A A Cognito user pool for sign-in and social federation
- ☐ B A Cognito identity pool issuing temporary credentials mapped to a scoped IAM role
- ☐ C An IAM user created for each registered customer
- ☐ D A publicly writable S3 bucket with post-upload validation

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- A

 User pool
Handles sign-up, sign-in and social federation for customer identity at this scale.
- B

 Identity pool with scoped role
Exchanges the token for temporary AWS credentials so the browser writes to S3 directly, with policy variables scoping each user to their own prefix.
- C

 IAM user per customer
IAM is for workforce and workloads, and its quotas are nowhere near five million.
- D

 Public bucket
Allows anyone to write anything. Validating afterwards does not prevent the damage.

THE SIGNAL

Both halves of Cognito in one question. Authentication is the user pool; AWS credentials are the identity pool.

A security team must be paged automatically when the string 'root login' appears more than three times in ten minutes across an application's CloudWatch log group.

Nobody should have to run a query to discover this.

Which approach meets this requirement?

- ☐ A A CloudWatch Logs metric filter with an alarm and SNS
- ☐ B A CloudWatch Logs Insights query run on a schedule
- ☐ C Amazon Macie scanning the log group
- ☐ D AWS Config with a custom rule

◆ PAUSE AND DECIDE

Which approach meets this requirement?

- A

Metric filter, alarm, SNS
The filter turns the log pattern into a metric, the alarm applies the count and window, and SNS pages the team.
- B

Scheduled Insights query
Insights returns results to a person. Wrapping it in scheduling and alerting recreates what a metric filter already does.
- C

Macie
Finds sensitive data inside S3 objects. It does not watch CloudWatch log groups.
- D

Config custom rule
Evaluates resource configuration, not the contents of log streams.

THE SIGNAL

Automated alerting on a log pattern is a metric filter. Insights cannot trigger anything.

Employees must sign in to a new internal web application on EC2 behind an Application Load Balancer using their existing corporate identity provider.

The team does not want to implement authentication logic in the application, and does not want a second set of passwords.

Which approach meets this requirement?

- ☐ A ALB authentication against a Cognito user pool federated to the corporate provider
- ☐ B IAM users for each employee with access keys
- ☐ C A Cognito identity pool with guest access enabled
- ☐ D Employee passwords stored in AWS Secrets Manager

◆ PAUSE AND DECIDE

Which approach meets this requirement?

- A

 ALB auth via federated user pool
The load balancer authenticates before traffic reaches the instances, and federation means no new passwords exist.
- B

 IAM users
IAM controls access to AWS APIs, not sign-in to your own web application.
- C

 Guest access
Unauthenticated access is the opposite of a sign-in requirement.
- D

 Passwords in Secrets Manager
Creates the second password set the team wants to avoid, and it is not an authentication service.

THE SIGNAL

An ALB can authenticate through Cognito or OIDC itself, so the application implements no sign-in logic.

Two thousand EC2 instances across fourteen accounts must be patched monthly with evidence of compliance.

No inbound ports may be opened and no SSH keys distributed. Instances run both Linux and Windows.

Which solution meets these requirements?

- ☐ A AWS Systems Manager Patch Manager with maintenance windows
- ☐ B A bastion host running a configuration management tool monthly
- ☐ C AWS Config rules reporting on patch compliance
- ☐ D Rebuilding every instance monthly from a fresh AMI

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

 Patch Manager
Agent-based so no inbound ports or keys, covers both operating systems and many accounts, and reports patch compliance.
- B

 Bastion and config tool
Requires exactly the open ports and distributed keys the question forbids.
- C

 Config rules
Reports whether patching happened. It does not patch anything.
- D

 Monthly rebuild
An enormous undertaking for two thousand mixed instances, with no compliance reporting.

THE SIGNAL

No ports, no keys, fleet-wide action means Systems Manager. Config reports; it never acts.

An insurer receives sixty thousand scanned claim forms monthly with varying layouts. It needs the policy number, claim value and claimant name loaded into a database.

The team will not train a machine learning model.

Which service should be used?

- ☐ A Amazon Rekognition
- ☐ B Amazon Textract
- ☐ C Amazon Comprehend
- ☐ D Amazon SageMaker with a custom model

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Rekognition
Detects objects, faces and text in images, but does not understand form structure or key-value relationships.
- C

 Comprehend
Analyses text that already exists. It cannot read a scanned image.
- B

 Textract
Built to extract text, tables and form key-value pairs from scanned documents, including unfamiliar layouts.
- D

 Custom SageMaker model
Requires the training and labelling the question explicitly excludes.

THE SIGNAL

Rekognition sees images. Textract reads documents. Structured form extraction is always Textract.

Regulation requires encryption keys to be generated and held in dedicated single-tenant hardware under the company's exclusive control.

Shared key infrastructure is not permitted.

Which service should be used?

- ☐ A AWS KMS with a customer managed key
- ☐ B AWS KMS with imported key material
- ☐ C AWS CloudHSM
- ☐ D AWS Secrets Manager

◆ PAUSE AND DECIDE

Which service should be used?

- A

 KMS customer managed key
Gives you policy control, but KMS hardware is multi-tenant, which the regulation forbids.
- B

 KMS imported material
Lets you originate the key, but it still resides in shared KMS infrastructure.
- C

 CloudHSM
Dedicated single-tenant hardware security modules under your exclusive control.
- D

 Secrets Manager
Stores and rotates secrets. It is not key management hardware.

THE SIGNAL

Single-tenant hardware or a named FIPS validation level is the one CloudHSM signal.

An on-call team is paged whenever any single instance in a large Auto Scaling group breaches its CPU alarm, even though the group replaces it automatically and the service stays healthy.

They want paging only when high CPU coincides with the load balancer reporting unhealthy targets.

What should be configured?

- ☐ A A composite alarm combining both alarms with AND
- ☐ B A higher CPU alarm threshold
- ☐ C More evaluation periods on the CPU alarm
- ☐ D Notifications routed to an unsubscribed SNS topic

◆ PAUSE AND DECIDE

What should be configured?

- A

 Composite alarm with AND
Fires only when both conditions hold, which is exactly what composite alarms exist for.
- B

 Higher threshold
Reduces noise by also missing genuine incidents.
- C

 More evaluation periods
Delays every alert including real ones, and a sustained single-instance breach still pages.
- D

 Unsubscribed topic
Silences the alert rather than making it meaningful.

THE SIGNAL

'Only when two things are both true' is the composite alarm signature.

A checkout page was broken for fifty minutes overnight by a faulty third-party script. Instances were healthy, CloudWatch metrics were normal, and the server returned pages successfully.

Nobody noticed until customers complained.

Which solution would detect this in future?

- ☐ A A CloudWatch Synthetics canary running the checkout journey
- ☐ B A CloudWatch alarm on load balancer request count
- ☐ C Detailed monitoring on the web instances
- ☐ D A metric filter on the web server access logs

◆ PAUSE AND DECIDE

Which solution would detect this in future?

- A

 Synthetics canary
Scripts the real user journey from outside and fails when the journey fails, regardless of infrastructure health.
- B

 Request count alarm
Requests were arriving and being served throughout. That metric looked normal.
- C

 Detailed monitoring
More frequent infrastructure metrics, which were never the failing signal.
- D

 Access log metric filter
The server returned pages successfully. The breakage was client-side.

THE SIGNAL

Infrastructure healthy but the user journey broken is the canary signature.

A disaster recovery plan claims a three-hour RTO based on restoring snapshots into a second Region. It has never been tested, and a tabletop exercise found the templates reference resources that do not exist there.

Nobody knows how long a restore actually takes.

What should the solutions architect recommend FIRST?

- ☐ A Upgrade to warm standby
- ☐ B Run a full DR test to measure the actual RTO and fix what it reveals
- ☐ C Copy the missing resources and consider the plan complete
- ☐ D Increase snapshot frequency

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Upgrade to warm standby
*Spends money avoiding a process nobody has measured.
The current plan may meet three hours once fixed.*
- B

 Test, measure, fix
*An untested plan has an unknown RTO. Testing validates it
and surfaces the faults the exercise hinted at.*
- C

 Fix and stop
*Repairs the one fault found by accident, strongly
suggesting others remain.*
- D

 More snapshots
Improves RPO, which nobody raised.

THE SIGNAL

An untested DR plan does not have an RTO, it has an aspiration.

A nightly report is triggered by a cron job on a small EC2 instance whose only purpose is to invoke a Lambda function at 03:00.

The instance was terminated accidentally and reports stopped silently for two weeks.

What should replace the cron job?

- ☐ A An EventBridge scheduled rule invoking the function
- ☐ B A second instance running the same cron job
- ☐ C An Auto Scaling group with a minimum of one instance
- ☐ D A CloudWatch alarm triggering the function

◆ PAUSE AND DECIDE

What should replace the cron job?

- A

 EventBridge scheduled rule
Managed scheduling with no instance at all, so there is nothing to terminate or patch.
- B

 Second instance
Two instances to maintain, and the job would run twice nightly.
- C

 Auto Scaling minimum one
Fixes the symptom while leaving you owning a server whose only job is running a clock.
- D

 Metric-triggered alarm
Alarms respond to metric conditions, not to a time of day.

THE SIGNAL

An instance whose only job is a schedule is always the wrong answer.

A company's backup software writes to a physical tape library reaching end of life. Replacing the software would need retraining and revalidation the business will not fund.

Backups should be stored durably in AWS instead.

Which solution should be used?

- ☐ A Storage Gateway in Tape Gateway mode
- ☐ B Storage Gateway in File Gateway mode
- ☐ C AWS Backup against the on-premises servers
- ☐ D AWS DataSync copying tape contents nightly

◆ PAUSE AND DECIDE

Which solution should be used?

- A

Tape Gateway
Presents a virtual tape library, so the existing software writes to what it believes is tape while data lands in S3 and Glacier.
- B

File Gateway
Presents a file share. Backup software expecting a tape library will not write to it unchanged.
- C

AWS Backup
Does not preserve the existing tape-based workflow.
- D

DataSync from tape
Moves files between file and object storage. It does not read a physical tape library.

THE SIGNAL

Existing backup software plus tape plus unwillingness to change is the one Tape Gateway signal.

Requests through a nine-service microservices chain occasionally take seven seconds while most finish in under one. Every individual service reports response times within its normal range.

The team cannot determine where the time is spent.

Which service should be used?

- ☐ A AWS X-Ray with tracing across the services
- ☐ B CloudWatch Logs Insights across all log groups
- ☐ C CloudWatch detailed monitoring on each service
- ☐ D VPC Flow Logs between the services

◆ PAUSE AND DECIDE

Which service should be used?

- A

 X-Ray
Follows a single request end to end across all nine services and attributes the time precisely.
- B

 Logs Insights
Queries each service's logs but cannot correlate one request across nine without a shared trace.
- C

 Detailed monitoring
More frequent averages, and averages are exactly what hides a slow minority of requests.
- D

 Flow Logs
Record which traffic flowed, not how long processing took.

THE SIGNAL

Every service healthy individually but the whole request slow is the distributed tracing signature.

An operations team wants alerts when memory utilisation exceeds eighty-five percent on a fleet of EC2 instances. They cannot find a memory metric in CloudWatch. They ask whether a third-party tool is required.

What should the solutions architect tell them?

- ☐ A Enable detailed monitoring to expose memory metrics
- ☐ B Install the CloudWatch agent to publish memory as a custom metric
- ☐ C Memory cannot be monitored without a third-party tool
- ☐ D Derive memory pressure from CPU credit balance

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

- A

 Detailed monitoring
Changes frequency from five minutes to one. It adds no new metric types, and memory is still absent.
- C

 Third-party required
An invented restriction. The agent does exactly this.
- B

 CloudWatch agent
Runs inside the operating system where memory is visible, and publishes it as a custom metric.
- D

 Infer from CPU credits
CPU credits relate to burstable instance performance, not memory.

THE SIGNAL

Detailed monitoring is about how often. The agent is about what.

A mobile app must push live updates to all connected devices within a second, let each screen request only the fields it needs, and work offline with synchronisation on reconnect. The team does not want to operate WebSocket servers.

Which service should be used?

- ☐ A AWS AppSync
- ☐ B API Gateway REST APIs with client polling
- ☐ C An ALB with EC2 WebSocket servers
- ☐ D Amazon SNS mobile push

◆ PAUSE AND DECIDE

Which service should be used?

- A

 AppSync
Managed GraphQL with real-time subscriptions, field-level queries and offline sync. All three requirements in one service.
- B

 REST with polling
Polling every second from every device scales badly, and REST returns fixed response shapes.
- C

 Self-managed WebSockets
Exactly the infrastructure the team said it will not run.
- D

 SNS mobile push
Delivers notifications. It is not a data API and offers no field selection or offline sync.

THE SIGNAL

Subscriptions plus field selection plus offline sync is the AppSync signature.

Factory control software must respond to equipment within five milliseconds, and regulation requires production data never to leave the site.

The company wants consistent AWS APIs across its estate.

Which solution meets these requirements?

- ☐ A AWS Outposts in the company's data centre
- ☐ B An AWS Local Zone in the nearest city
- ☐ C AWS Wavelength in the local 5G network
- ☐ D The nearest Region with Direct Connect

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

Outposts
AWS-managed racks physically on site, so data never leaves and latency to local equipment is minimal, with the same APIs.
- B

Local Zone
Closer than a Region but still off site, so the data residency requirement fails.
- C

Wavelength
For mobile devices on a 5G network, not wired factory equipment.
- D

Region plus Direct Connect
Data leaves the site and the round trip cannot reliably hit five milliseconds.

THE SIGNAL

Local Zones and Wavelength are AWS facilities nearer you. Only Outposts is on your premises.

A contact centre must produce searchable transcripts of recorded calls and identify which calls contained customer frustration.

The company will not build or train models.

Which combination meets these requirements? (Choose two.)

- | | |
|--|---|
| ☐ A Amazon Transcribe on the recordings | ☐ B Amazon Comprehend on the transcripts |
| ☐ C Amazon Polly on the recordings | ☐ D Amazon Rekognition on the recordings |

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- ☒ A Transcribe
Converts recorded audio into the searchable text transcripts required.
- ☒ B Comprehend
Runs sentiment analysis over the transcripts to identify frustration.
- ☐ C Polly
Turns text into speech, which is the opposite direction.
- ☐ D Rekognition
Analyses images and video. These are audio recordings.

THE SIGNAL

Audio to text with Transcribe, then text to meaning with Comprehend. Polly runs the other way.

A company plans to migrate five hundred servers within five months. Applications are stable, the business will not fund code changes during the move, and speed is the priority.

A consultant proposes refactoring everything to containers as part of the migration.

What should the solutions architect advise?

- ☐ A Rehost with AWS Application Migration Service, and modernise afterwards
- ☐ B Proceed with refactoring, since containers are the modern architecture
- ☐ C Repurchase SaaS equivalents for all five hundred applications
- ☐ D Retain everything on premises

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Rehost now, modernise later
Fastest path with no application changes, and it does not prevent refactoring once workloads run on AWS.
- B

 Refactor during migration
The slowest strategy, needing exactly the code changes the business has ruled out.
- C

 Repurchase everything
Replacing five hundred applications in five months is not achievable, and would change every business process.
- D

 Retain on premises
Ignores the requirement to migrate.

THE SIGNAL

Speed plus no code changes means rehost. Refactoring is not wrong generally, only under these constraints.

During a migration assessment of six hundred servers, eighty are found running applications no department claims, with no logins in two years and no traffic beyond monitoring. The migration budget is under pressure.

Which strategy applies to those eighty servers?

- ☐ A Retire them
- ☐ B Rehost them onto the smallest instances available
- ☐ C Retain them on premises until an owner appears
- ☐ D Replatform them onto managed services

◆ PAUSE AND DECIDE

Which strategy applies to those eighty servers?

- | | |
|---|--|
| <p>A Retire</p> <p>No owner, no logins in two years, no traffic. Retiring removes over thirteen percent of the migration scope and its ongoing cost.</p> | <p>B Rehost small</p> <p><i>Spends migration effort and ongoing money running software nobody uses.</i></p> |
| <p>C Retain</p> <p><i>Keeps paying for hardware and licences for unused software, and defers the decision.</i></p> | <p>D Replatform</p> <p><i>Invests engineering effort modernising applications nobody uses.</i></p> |

THE SIGNAL

Retire is a real strategy. Finding things to switch off is a main reason to run an assessment.

A team stores application logs from four hundred instances in CloudWatch Logs. Log groups were created years ago and retention was never configured.

The finance team has queried a steadily growing CloudWatch bill. Audit requires twelve months of logs.

Which combination is MOST cost-effective? (Choose two.)

- ☐ A Set a twelve-month retention period on the log groups
- ☐ B Export logs older than the queryable window to S3
- ☐ C Delete the log groups and write logs to local disk instead
- ☐ D Enable detailed monitoring to reduce log volume

◆ PAUSE AND DECIDE

Which combination is MOST cost-effective? (Choose two.)

- A

 Set retention
CloudWatch Logs retention defaults to never expire, so unbounded data has been accumulating charges for years.
- B

 Export older logs to S3
S3 is considerably cheaper for long-term retention while still satisfying the audit requirement.
- C

 Local disk
Logs are lost when instances are replaced, and nothing is centralised or queryable.
- D

 Detailed monitoring
Concerns metric frequency and increases cost. It has nothing to do with log volume.

THE SIGNAL

The default retention is 'never expire'. Setting it is one of the easiest savings on any AWS bill.

A retailer wants personalised product recommendations from three years of customer browsing and purchase history.

The team has no machine learning specialists and wants results within weeks.

Which service should be used?

- | | |
|---|--|
| ☐ A Amazon Personalize | ☐ B Amazon Forecast |
| ☐ C Amazon SageMaker with a custom model | ☐ D Amazon Comprehend |

◆ PAUSE AND DECIDE

Which service should be used?

- | | |
|--|--|
| <p>A Personalize</p> <p>Built to produce recommendations from interaction history with no ML expertise required.</p> | <p>B Forecast</p> <p><i>Predicts future values of a time series such as demand. That is forecasting, not personalisation.</i></p> |
| <p>C Custom SageMaker model</p> <p><i>Needs the specialists the team lacks, on a timeline that does not allow it.</i></p> | <p>D Comprehend</p> <p><i>Analyses text for sentiment and entities. It does not generate recommendations.</i></p> |

THE SIGNAL

Personalize recommends to a person. Forecast predicts a quantity over time. Both use history.

Scoring Practice Exam 5

This paper is weighted towards the newer material, so compare it against Exam 4

specifically

- ◆ **Similar to your Exam 4 score.** The monitoring, migration and identity content has landed. You are ready.
- ◆ **Noticeably lower than Exam 4.** The gap is in modules 13 to 15 rather than the four domains. Rewatch those three.
- ◆ **Higher than Exam 4.** Likely familiarity with recent material. Trust the average of all five, not the best one.
- ◆ **Below 12 here but comfortable elsewhere.** Cognito and the managed AI services are the usual culprits. They are pure recognition, so they reward one focused pass.

ON THE EXAM

Five papers, one hundred original questions. Take the average of your last three as the honest estimate.